01

MODEL ROUTING ARCHITECTURE

How AI Systems Choose the Right Model for Every Task

RIGHT MODEL	LOWER COST	BETTER QUALITY	FASTER RESPONSES
for every task	Smarter utilization	Higher accuracy	Optimized performance

The smartest AI systems don't use one model.
They choose the right one.

VANSH BUDHIRAJA

02

WHY MODEL ROUTING?

The **right model** for the **right task** makes all the difference.

WITHOUT ROUTING
One model handles everything

User

↓

One LLM
(Used for all tasks)

↓

All Tasks

- Coding
- Reasoning
- Translation
- Vision
- Math
- More...

SLOW
Not optimized for every task.

EXPENSIVE
High cost even for simple tasks.

LOWER QUALITY
Suboptimal performance overall.

VS

WITH MODEL ROUTING
Router sends each task to the best model

User

↓

Routing Engine
(Smart Decision Layer)

Code Generation	→	**Claude 3.5** (Better for code)
Complex Reasoning	→	**GPT-5** (Stronger reasoning)
Translation	→	**Gemini 1.5** (Better at languages)
Image Understanding	→	**GPT-4o** (Best for vision)
Simple / Fast Tasks	→	**Llama 3** (Fast + Efficient)

FASTER
Optimized models = quicker responses.

COST EFFICIENT
Right model = lower cost.

HIGHER QUALITY
Best model = better results.

Routing ensures every request reaches the best runway.

THE ROUTING DECISION PROCESS

From request to the best model in a few smart decisions.

USER REQUEST
e.g., "Explain quantum computing"

TASK DETECTION
Understand the intent and context

- CODE RELATED? — YES
- COMPLEX REASONING? — YES
- VISION / IMAGE TASK? — YES
- MATH / CALCULATION? — YES
- NEED REAL-TIME INFORMATION? — YES

CHOOSE THE BEST MODEL
Based on task type, context, cost, latency & performance

ROUTE TO THE MOST SUITABLE MODEL

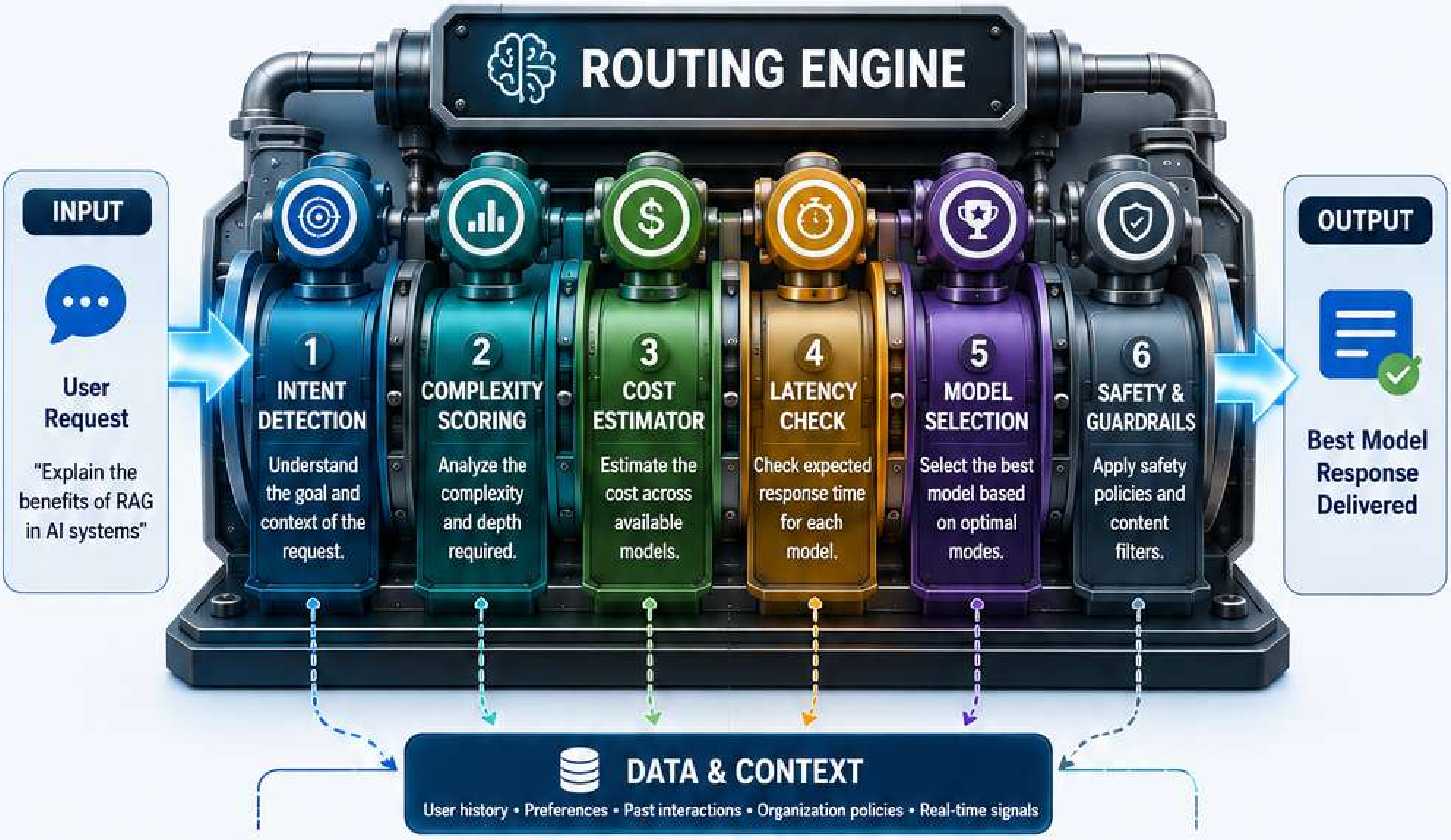
CODE GENERATION	COMPLEX REASONING	VISION / IMAGE	MATH / CALCULATIONS	REAL-TIME INFO
Claude 3.5 / GPT-4o	GPT-5	GPT-4o / Claude 3.5	Gemini 1.5 / GPT-5	Perplexity / DeepSeek
Best for coding, APIs & structured output	Best for deep reasoning, analysis & problem solving	Best for image understanding & multimodal tasks	Best for math, logic & numerical computations	Best for search, news, facts & real-time data

Every request is unique.
Route smart. Respond better.

04

INSIDE THE ROUTING ENGINE

Multiple smart modules work together to pick the perfect model.



WHAT EACH MODULE DOES

1. INTENT DETECTION
Detects what the user really wants to achieve.

Example: "Explain" → Informational

2. COMPLEXITY SCORING
Scores the request based on depth, logic and steps.

Example: Medium Complexity

3. COST ESTIMATOR
Predicts cost for each model based on tokens.

Example: Claude 3.5 → $0.02; GPT-4o → $0.06

4. LATENCY CHECK
Estimates response time from each model endpoint.

Example: Gemini 1.5 → 1.2s; GPT-4o → 2.3s

5. MODEL SELECTION
Chooses the model with the best overall score.

Example: Winner → Claude 3.5

6. SAFETY & GUARDRAILS
Ensures the response is safe, compliant and reliable.

Example: PII Check ✓; Bias Check ✓

KEY FACTORS CONSIDERED

Accuracy • Cost • Latency • Capacity & Load • Reliability • Safety & Compliance • User & Business Priorities

A smart engine. Many checks. One perfect decision.

Right model. Right time. Every time.

05

ROUTING STRATEGIES

Different strategies. Different goals. One smart router.

01 RULE-BASED ROUTING

Uses predefined rules based on conditions and parameters.

HOW IT WORKS

Request → If / Else Rules → Model A / Model B

Best for: Simple, predictable and structured scenarios.

02 INTENT-BASED ROUTING

Understands user intent and routes to the model best suited for that intent.

HOW IT WORKS

Request → Intent Detection → Code / Reasoning / Translate

Code → Model A

Reasoning → Model B

Translate → Model C

Best for: Natural language requests with varied intents.

03 COST-BASED ROUTING

Chooses the most cost-effective model that meets quality needs.

HOW IT WORKS

Request → Cost Estimator → Pick Lowest Cost Model (Quality OK) → Selected Model

Best for: Cost optimization at scale.

04 PERFORMANCE ROUTING

Routes to the model with the best accuracy or quality score.

HOW IT WORKS

Request → Evaluate Models (Quality Score) → Pick Highest Score Model → Selected Model

Best for: Quality-critical and high-stakes tasks.

05 LATENCY-BASED ROUTING

Selects the model that provides the fastest response time.

HOW IT WORKS

Request → Check Latency (Real-time) → Pick Lowest Latency Model → Selected Model

Best for: Real-time applications and user experience.

06 HYBRID ROUTING

Combines multiple strategies for the best overall decision.

HOW IT WORKS

Request → Intent / Cost / Latency / Quality → Weighted Decision Engine → Best Model Selected

Best for: Complex enterprise environments.

QUICK COMPARISON

STRATEGY	Rule-Based	Intent-Based	Cost-Based	Performance	Latency-Based	Hybrid
FOCUS	Conditions	User Intent	Cost	Quality	Speed	Multi-Factor
TRADE-OFF	Less Flexible	Needs Good Intent Detection	May Lower Quality	Higher Cost	May Lower Quality	More Complex

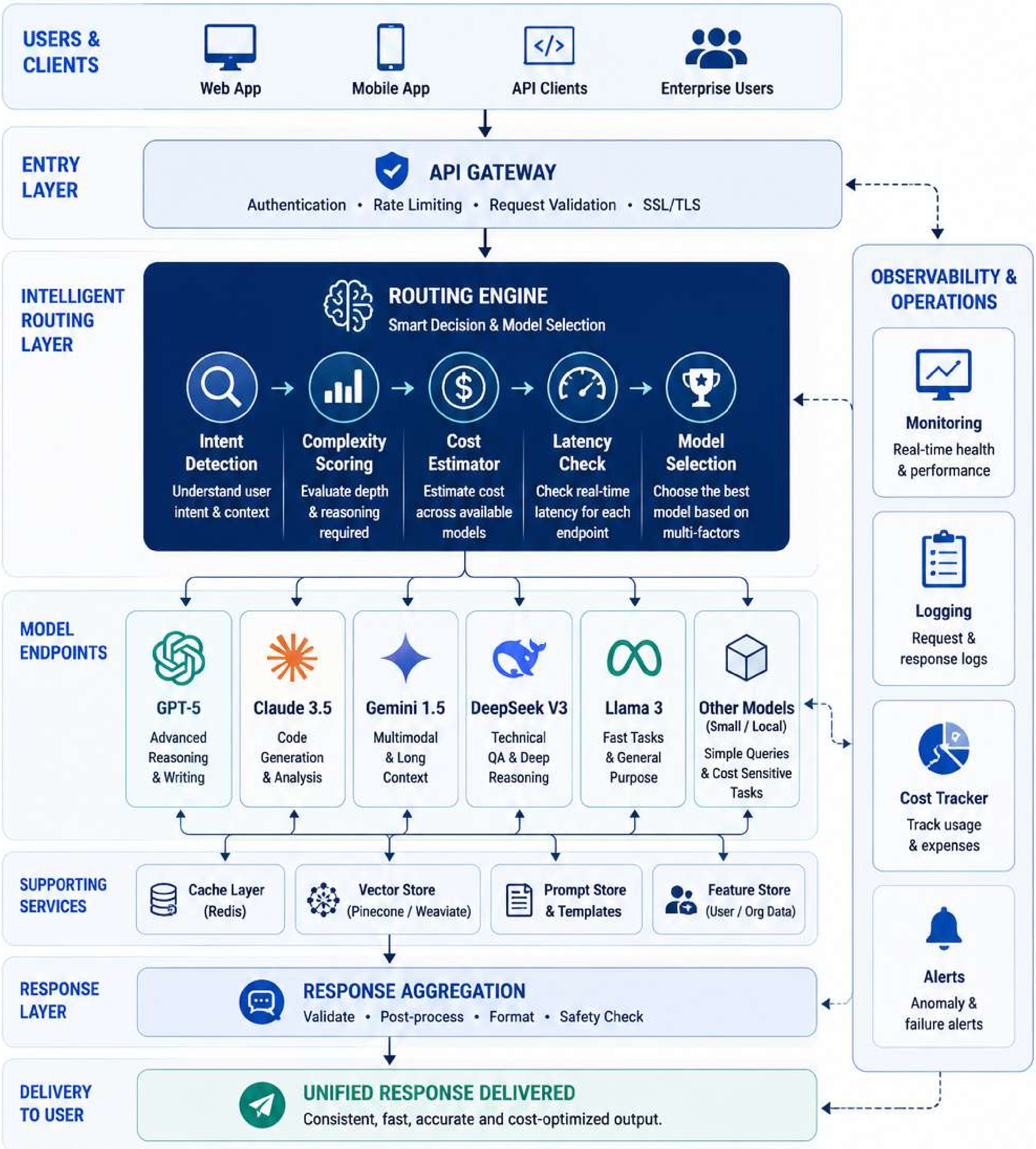
No one strategy fits all.

Smart routing uses the right strategy at the right time.

06

ENTERPRISE ROUTING ARCHITECTURE

A scalable architecture to route every request to the best model.



SMARTER DECISIONS
Right model for every request

LOWER COSTS
Optimized usage across models

FASTER RESPONSES
Reduced latency & improved UX

BETTER RELIABILITY
Monitored, secure & enterprise-ready

07

END-TO-END
ROUTING FLOW

From user request to the best response — seamlessly.

01 USER
User submits a request

"Explain the benefits of RAG in AI systems"

02 REQUEST
Request received by the system

Captured with context, history & metadata

03 ROUTING ENGINE
Analyzes the request using multiple factors

Intent • Complexity
Cost • Latency
Performance

04 BEST MODEL SELECTED
The most suitable model is chosen

Right model for the right task at the right time

05 VALIDATION & POST-PROCESS
Response validated and enhanced

Safety checks, formatting & quality control

06 FINAL RESPONSE
Deliver accurate, fast & optimized response

Returned to user seamlessly

FEEDBACK LOOP
Learn • Improve • Optimize

EXAMPLE ROUTING IN ACTION

User Request

"Write a Python function to calculate prime numbers"

Routing Engine

- Intent: Code Generation
- Complexity: Medium
- Cost Check: Low
- Latency: Fast
- Best Fit: Claude 3.5

Selected Model

Claude 3.5

(Best for coding tasks)

Final Response

WITHOUT ROUTING VS WITH ROUTING

WITHOUT ROUTING	WITH ROUTING
HIGHER COSTS Expensive models used for every request.	**LOWER COSTS** Right model for the task = optimized spend.
SLOWER RESPONSES Longer latency due to non-optimized selection.	**FASTER RESPONSES** Optimized latency with the best-suited model.
LOWER QUALITY Generic answers, not the best fit for the task.	**BETTER ANSWERS** Higher accuracy and better user satisfaction.

Smart routing. Smarter AI.
Right model. Right time. Better outcomes.

VANSH BUDHIRAJA